https://doi.org/10.1063/5.0255841

Phase behavior of Cacio e Pepe sauce

G. Bartolucci; D. M. Busiello; M. Ciarchi; A. Corticelli; I. Di Terlizzi; F. Olmeda; D. Revignas; V. M. Schimmenti

Physics of Fluids
DOI: 10.1063/5.0255841

Abstract

“Pasta alla Cacio e pepe” is a traditional Italian dish made with pasta, pecorino cheese, and pepper. Despite its simple ingredient list, achieving the perfect texture and creaminess of the sauce can be challenging. In this study, we systematically explore the phase behavior of Cacio e pepe sauce, focusing on its stability at increasing temperatures for various proportions of cheese, water, and starch. We identify starch concentration as the key factor influencing sauce stability, with direct implications for practical cooking. Specifically, we delineate a regime where starch concentrations below 1% (relative to cheese mass) lead to the formation of system-wide clumps, a condition determining what we term the “Mozzarella Phase” and corresponding to an unpleasant and separated sauce. Additionally, we examine the impact of cheese concentration relative to water at a fixed starch level, observing a lower critical solution temperature that we theoretically rationalized by means of a minimal effective free-energy model. We further analyze the effect of a less traditional stabilizer, trisodium citrate, and observe a sharp transition from the Mozzarella Phase to a completely smooth and stable sauce, in contrast to starch-stabilized mixtures, where the transition is more gradual. Finally, we present a scientifically optimized recipe based on our findings, enabling a consistently flawless execution of this classic dish.

Metadati bibliografici e abstract originali del record pubblico dell’editore.